

Complete EKF Derivation for Indoor Navigation System (RSSI + IMU Fusion)

1. Problem Setup

4 ESP anchor nodes at known coordinates.

Robot has ESP32 + MPU-9250.

Anchors -> RSSI -> distance estimate -> trilateration -> (x,y).

IMU gives acceleration, gyro, magnetometer heading.

EKF fuses all.

2. State Vector

$$X = [x, y, v, \theta, \omega]^T$$

$x, y = \text{position}$

$v = \text{forward linear velocity}$

$\theta = \text{heading (yaw)}$

$\omega = \text{angular velocity}$

3. Continuous Model

$$\frac{dx}{dt} = v \cos(\theta)$$

$$\frac{dy}{dt} = v \sin(\theta)$$

$$\frac{dv}{dt} = a$$

$$\frac{d\theta}{dt} = \omega$$

$$\frac{d\omega}{dt} = 0$$

Angular velocity derivative assumed zero because over small timestep Δt angular rate changes little; unmodeled change enters process noise.

4. Euler Discretization

General:

$$\begin{aligned} \text{If } \frac{ds}{dt} &= f(s), \\ \text{then } s(k+1) &= s(k) + \left(\frac{ds}{dt}\right) \Delta t \end{aligned}$$

Applying:

$$x(k+1) = x(k) + v(k) \cos(\theta(k)) \Delta t$$

$$y(k+1) = y(k) + v(k) \sin(\theta(k)) \Delta t$$

$$v(k+1) = v(k) + a(k) \Delta t$$

$$\theta(k+1) = \theta(k) + \omega(k) \Delta t$$

$$\omega(k+1) = \omega(k)$$

5. Process Model

$$X(k+1) = f(X_k, u_k) + w_k$$

6. Jacobian

Needed because system is nonlinear (sin/cos terms).

$$F = \frac{\partial f}{\partial X}$$

Rows derived from partial derivatives:

$$[1, 0, \cos(\theta) \Delta t, -v \sin(\theta) \Delta t, 0]$$

$$[0, 1, \sin(\theta) \Delta t, v \cos(\theta) \Delta t, 0]$$

$$[0, 0, 1, 0, 0]$$

$$[0, 0, 0, 1, \Delta t]$$

$$[0, 0, 0, 0, 1]$$

7. Covariance

P is 5x5 uncertainty matrix.

Diagonal = variances of x, y, v, theta, omega.

8. Covariance Prediction

$$P^- = F P F^T + Q$$

Q = process noise covariance.

9. Measurement Vector

$$z = [x_{rssi}, y_{rssi}, \theta_{mag}]^T$$

Measurement function:

$$h(X) = [x, y, \theta]^T$$

10. Residual

$$y = z - h(X^-)$$

11. Measurement Jacobian

$$H = \begin{bmatrix} 1,0,0,0,0 \\ 0,1,0,0,0 \\ 0,0,0,1,0 \end{bmatrix}$$

12. Measurement Noise

$$R = \text{diag}(\sigma x^2, \sigma y^2, \sigma \theta^2)$$

13. Innovation Covariance

$$S = H P^- H^T + R$$

14. Kalman Gain

$$K = P^- H^T S^{-1}$$

15. State Update

$$X = X^- + K y$$

16. Covariance Update

$$P = (I - KH)P^-$$